LAPORAN TUGAS PRAKTIK KOMPREHENSIF - STRUKTUR SISTEM OPERASI

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PROGRAM STUDI TEKNIK INFORMATIKA

FAKULTAS TEKNIK

UNIVERSITAS MUHAMMADIYAH JEMBER

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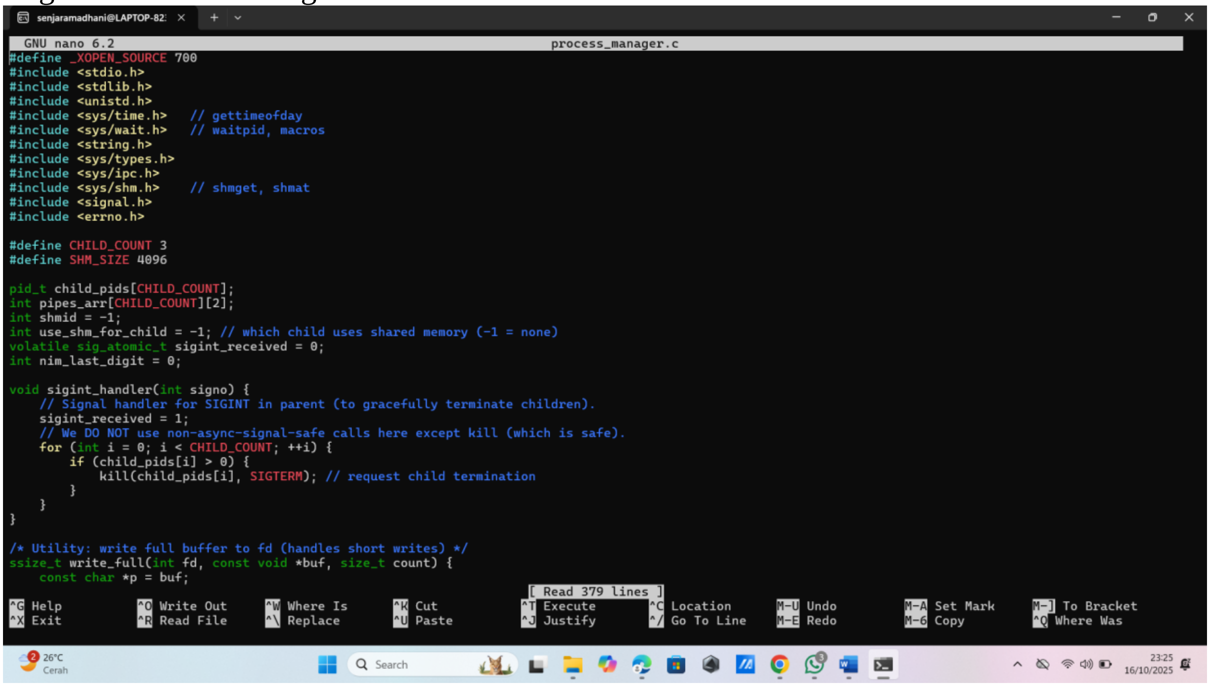
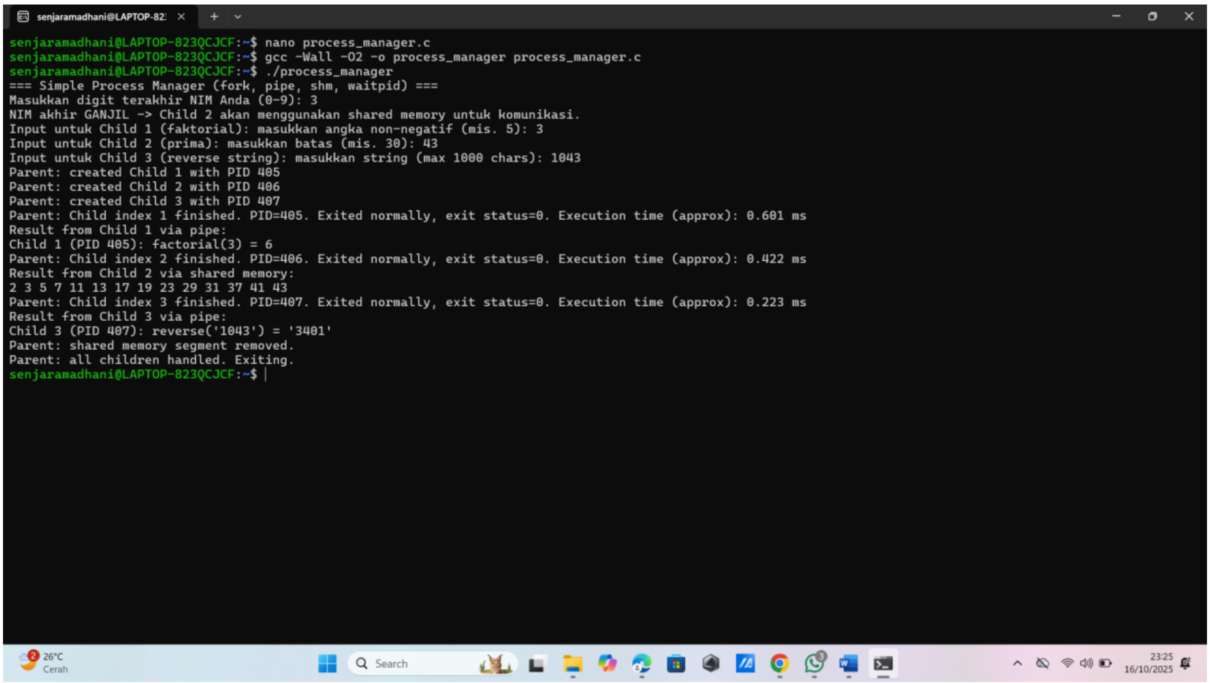
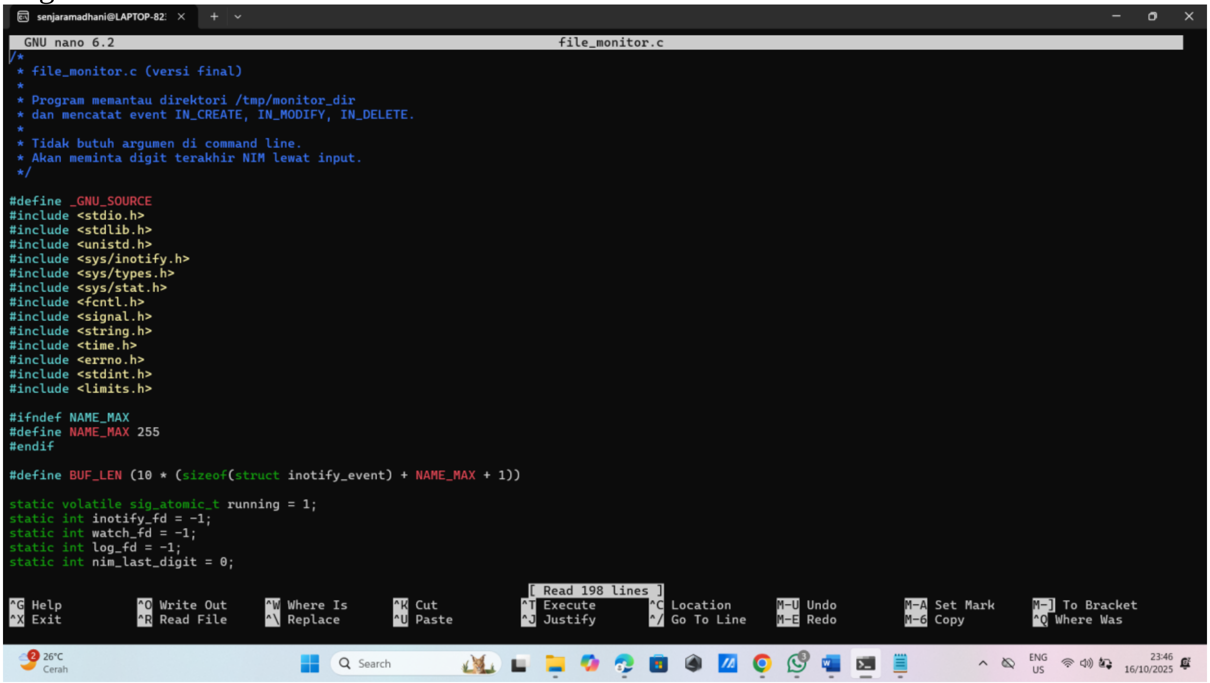
Tugas Praktik Komprehensif

Tugas 1: Process Manager

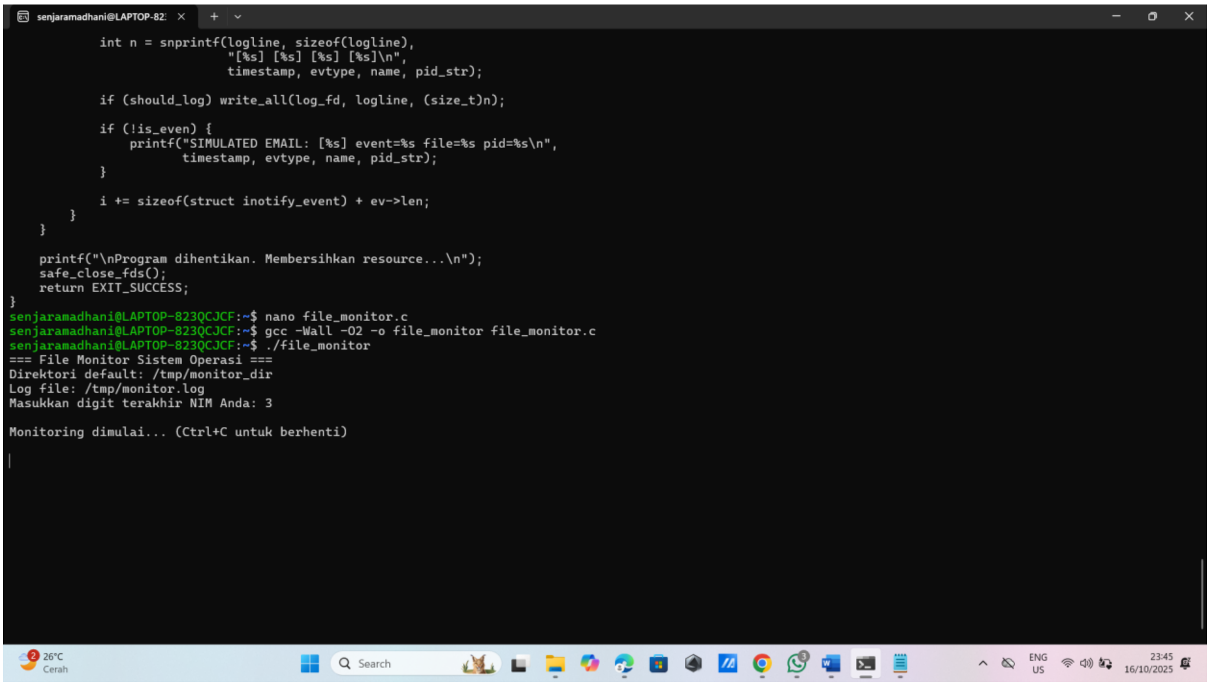




Tugas 2: File Monitor

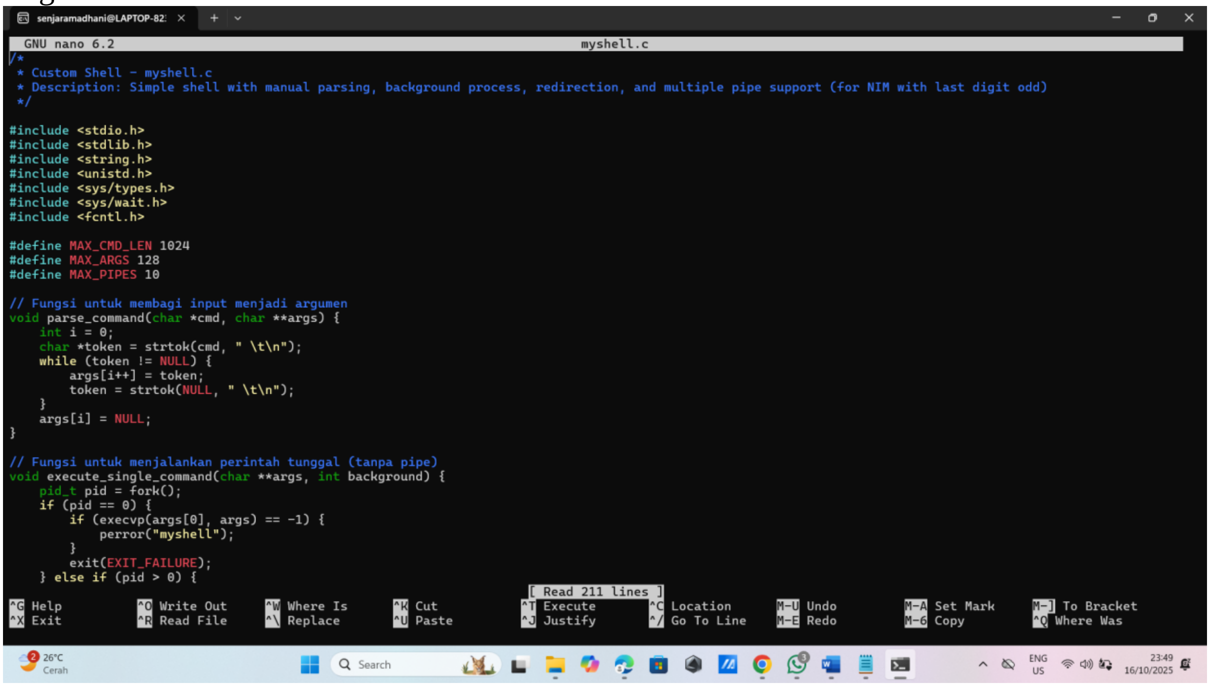


```
GNU nano 6.2                                file_monitor.c
/*
 * file_monitor.c (versi final)
 *
 * Program memantau direktori /tmp/monitor_dir
 * dan mencatat event IN_CREATE, IN_MODIFY, IN_DELETE.
 *
 * Tidak butuh argumen di command line.
 * Akan meminta digit terakhir NIM lewat input.
 */

#define _GNU_SOURCE
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/inotify.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>
#include <signal.h>
#include <string.h>
#include <time.h>
#include <errno.h>
#include <stdint.h>
#include <limits.h>

#ifndef NAME_MAX
#define NAME_MAX 255
#endif

#define BUF_LEN (10 * (sizeof(struct inotify_event) + NAME_MAX + 1))

static volatile sig_atomic_t running = 1;
static int inotify_fd = -1;
static int watch_fd = -1;
static int log_fd = -1;
static int nim_last_digit = 0;

                              [ Read 198 lines ]
^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy      ^Q Where Was
```

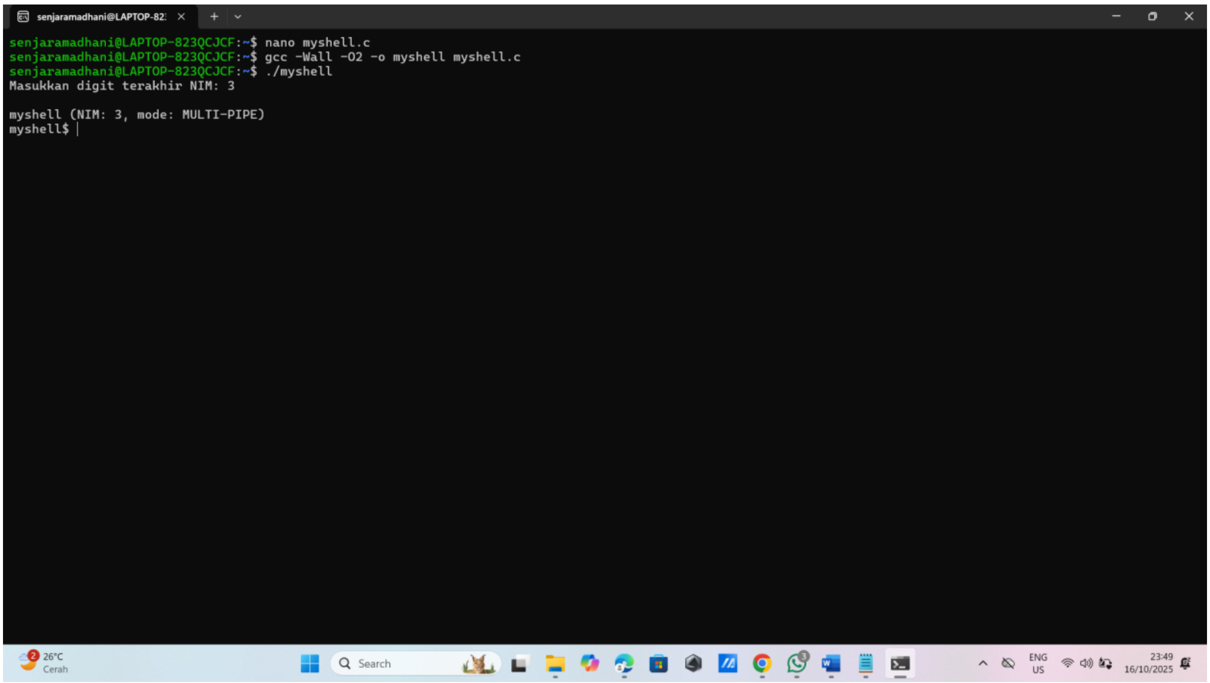


```
            int n = snprintf(logline, sizeof(logline),
                             "[%s] [%s] [%s] [%s]\n",
                             timestamp, evtype, name, pid_str);

            if (should_log) write_all(log_fd, logline, (size_t)n);

            if (!is_even) {
                printf("SIMULATED EMAIL: [%s] event=%s file=%s pid=%s\n",
                       timestamp, evtype, name, pid_str);
            }

            i += sizeof(struct inotify_event) + ev->len;
        }
    }

    printf("\nProgram dihentikan. Membersihkan resource...\n");
    safe_close_fds();
    return EXIT_SUCCESS;
}
senjaramadhani@LAPTOP-823QCJCF:~$ nano file_monitor.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -Wall -O2 -o file_monitor file_monitor.c
senjaramadhani@LAPTOP-823QCJCF:~$ ./file_monitor
=== File Monitor Sistem Operasi ===
Direktori default: /tmp/monitor_dir
Log file: /tmp/monitor.log
Masukkan digit terakhir NIM Anda: 3

Monitoring dimulai... (Ctrl+C untuk berhenti)
```

Tugas 3: Custom Shell



```
GNU nano 6.2                              myshell.c
/*
 * Custom Shell - myshell.c
 * Description: Simple shell with manual parsing, background process, redirection, and multiple pipe support (for NIM with last digit odd)
 */

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <fcntl.h>

#define MAX_CMD_LEN 1024
#define MAX_ARGS 128
#define MAX_PIPES 10

// Fungsi untuk membagi input menjadi argumen
void parse_command(char *cmd, char **args) {
    int i = 0;
    char *token = strtok(cmd, " \t\n");
    while (token != NULL) {
        args[i++] = token;
        token = strtok(NULL, " \t\n");
    }
    args[i] = NULL;
}

// Fungsi untuk menjalankan perintah tunggal (tanpa pipe)
void execute_single_command(char **args, int background) {
    pid_t pid = fork();
    if (pid == 0) {
        if (execvp(args[0], args) == -1) {
            perror("myshell");
        }
        exit(EXIT_FAILURE);
    } else if (pid > 0) {
                              [ Read 211 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark   M-] To Bracket
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy       ^Q Where Was
```



```
senjaramadhani@LAPTOP-823QCJCF:~$ nano myshell.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -Wall -O2 -o myshell myshell.c
senjaramadhani@LAPTOP-823QCJCF:~$ ./myshell
Masukkan digit terakhir NIM: 3

myshell (NIM: 3, mode: MULTI-PIPE)
myshell$ |
```

Tugas 4: System Information Tool